

HAMZA NAEEM JANJUA

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SUMMARY

Electrical Engineering undergraduate (CGPA 3.83/4.0) at NED University with hands-on experience across embedded systems, PCB design, digital logic, and applied programming. Has independently designed, built, and tested five end-to-end hardware/software projects spanning IoT, FPGA/ASIC-style digital design, and computer vision. Looking for an internship where I can contribute to real embedded systems or hardware design work.

EDUCATION

B.E. Electrical Engineering — NED University of Engineering & Technology 2024 – 2028 (Ongoing)

CGPA: 3.83 / 4.0 | Currently in Semester 5

Intermediate (Computer Science) — Government Dehli College 2022 – 2024

TECHNICAL SKILLS

Programming: C, C++, Python, Verilog, HTML, CSS

Tools: MATLAB, Arduino IDE, KiCad, Quartus, ModelSim, AutoCAD

Hardware: ESP32, ESP8266, microcontroller peripherals (I2C/SPI, sensors, displays)

Core Areas: PCB Design, Embedded Systems, Digital Logic Design, Circuit Analysis

PROJECTS

Smart Health Monitoring System (IoT)

- Built an ESP32-based vitals and environment monitor combining temperature, humidity, and heart-rate sensors with a live OLED display.
- Wrote firmware in C++ (Arduino framework) to poll sensors over I2C and refresh readings in real time without blocking the main loop.

Custom Arduino Nano PCB

- Designed a compact, single-board replacement for the Arduino Nano in KiCad, including schematic capture, footprint placement, and routing.
- Optimized component layout and trace routing for a reduced board footprint suitable for embedding in space-constrained enclosures.

5-Bit ALU (Verilog)

- Designed a 5-bit arithmetic logic unit in Verilog supporting core arithmetic and logic operations (add, subtract, AND, OR, shift).
- Built a self-checking testbench in ModelSim to simulate all operations and verify correct output against expected results.

Face Tracking System (Python + OpenCV)

- Developed a real-time face detection and tracking pipeline using OpenCV's Haar cascade / DNN-based detectors on live webcam input.
- Implemented frame-to-frame tracking logic to keep a stable lock on the detected face under movement.

IoT Temperature Logger (ESP8266)

- Built a cloud-connected temperature logging system on the ESP8266, pushing periodic sensor readings to a cloud dashboard over Wi-Fi.
- Implemented reconnect and error-handling logic to keep the device logging reliably through network drops.

Graph-Based Social Network Model

- Modeled a social network as a graph and implemented a BFS-based friend-recommendation algorithm in Python/C++.
- Tested the recommendation logic against sample networks of varying size and connectivity.

ACHIEVEMENTS

- Merit-based scholarship recipient, Classes 6–8, in recognition of consistent academic performance.

LEADERSHIP & ADDITIONAL INFORMATION

- Led and independently delivered multiple end-to-end academic hardware/software projects, from design through testing.